FraudGuard: Fake Job Listing Detection using Deep Learning and Agentic AI

Data Science and AI Lab Project – Final Presentation

Group: Group 9

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The Problem: Fake Job Listings at Scale

3–5% of all online job postings are fraudulent.
DATA METRIC

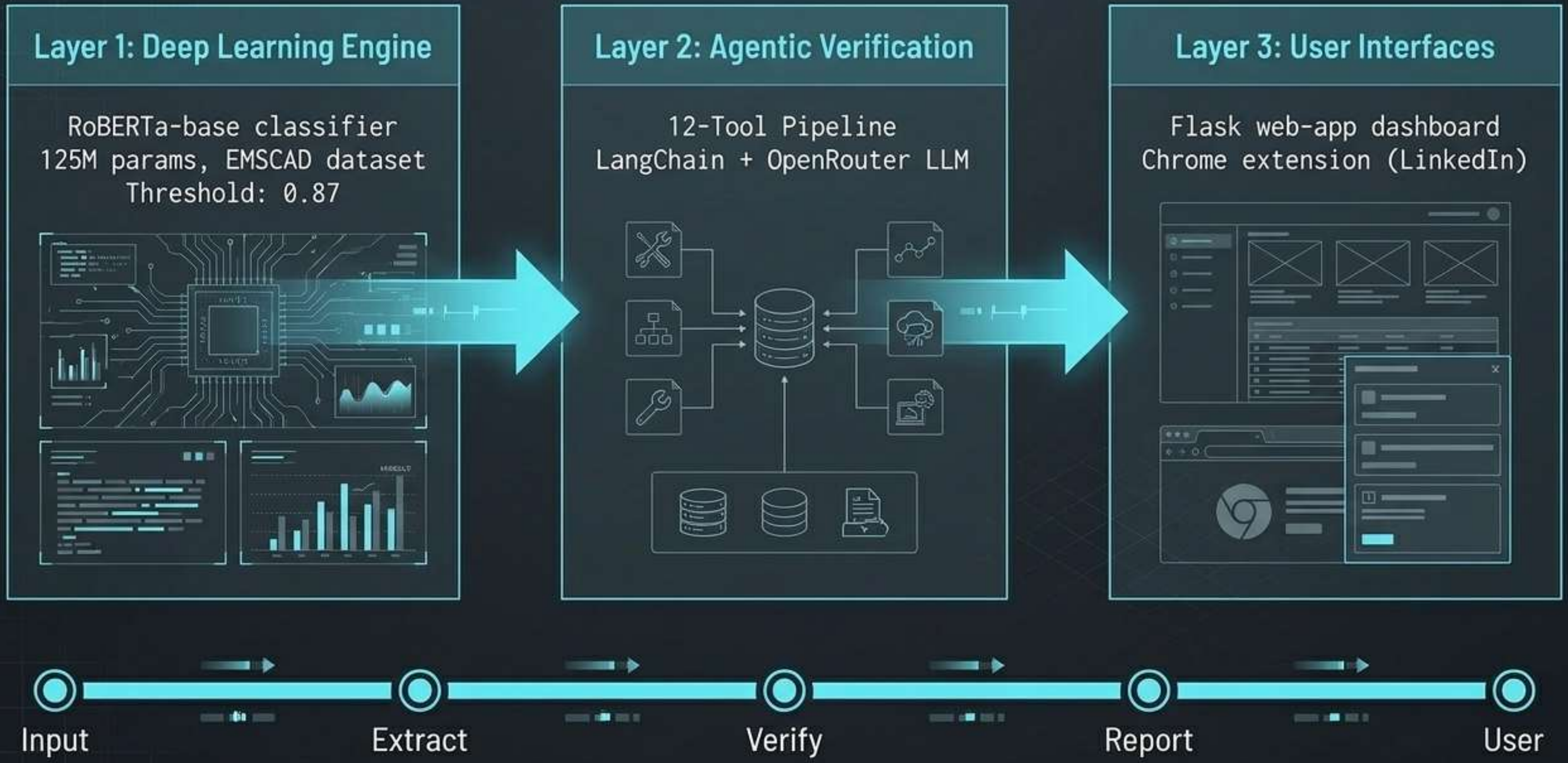
Modern fraud postings mimic legitimate ads with linguistic sophistication — keyword filters fail.

Victims lose money, personal data, and time — real-world impact on Indian job seekers.

No existing tool combines deep NLP + external verification + human-readable explanation.

FraudGuard doesn't just output a probability score; it **cross-references** the web to prove its verdict.

FraudGuard — Three-Layer Architecture



EMSCAD Dataset — 17,880 Job Postings

17,880 samples, 4.84% fraudulent (866 fraud, 17,014 legitimate) — 20:1 imbalance.

18 features: 5 free-text, 9 structured metadata, 3 binary.

After deduplication: 15,787 unique samples.

Split: 70% train / 15% validation / 15% test (stratified).

Preprocessing Innovation: The [SEP] Strategy

Structured metadata fields are placed FIRST in the [SEP]-joined input string.
This protects critical signals from 512-token truncation.

RoBERTa-base — Why and How

Architecture

roberta-base

12 transformer layers, 12 attention heads, 768-dim hidden state, 125.5M params

Classification head

Dropout(0.1) → Linear(768→2) → Softmax

Full fine-tuning (not LoRA)

small dataset needs all parameters to adapt

The Challenge: 20:1 Class Imbalance

Down-weights easy negatives, amplifies gradient signal on hard fraud examples.

Solution: Focal Loss (γ = 1.6920, fraud class weight = 2.8251)

25-Trial Optuna HPO — Systematic Search with Hard Floors

lr: 2.59e-5	warmup: 0.1506	batch: 16	weight_decay: 0.0702
focal_gamma: 1.6920	fraud_class_weight: 2.8251		epochs: 9 (stopped at 7)

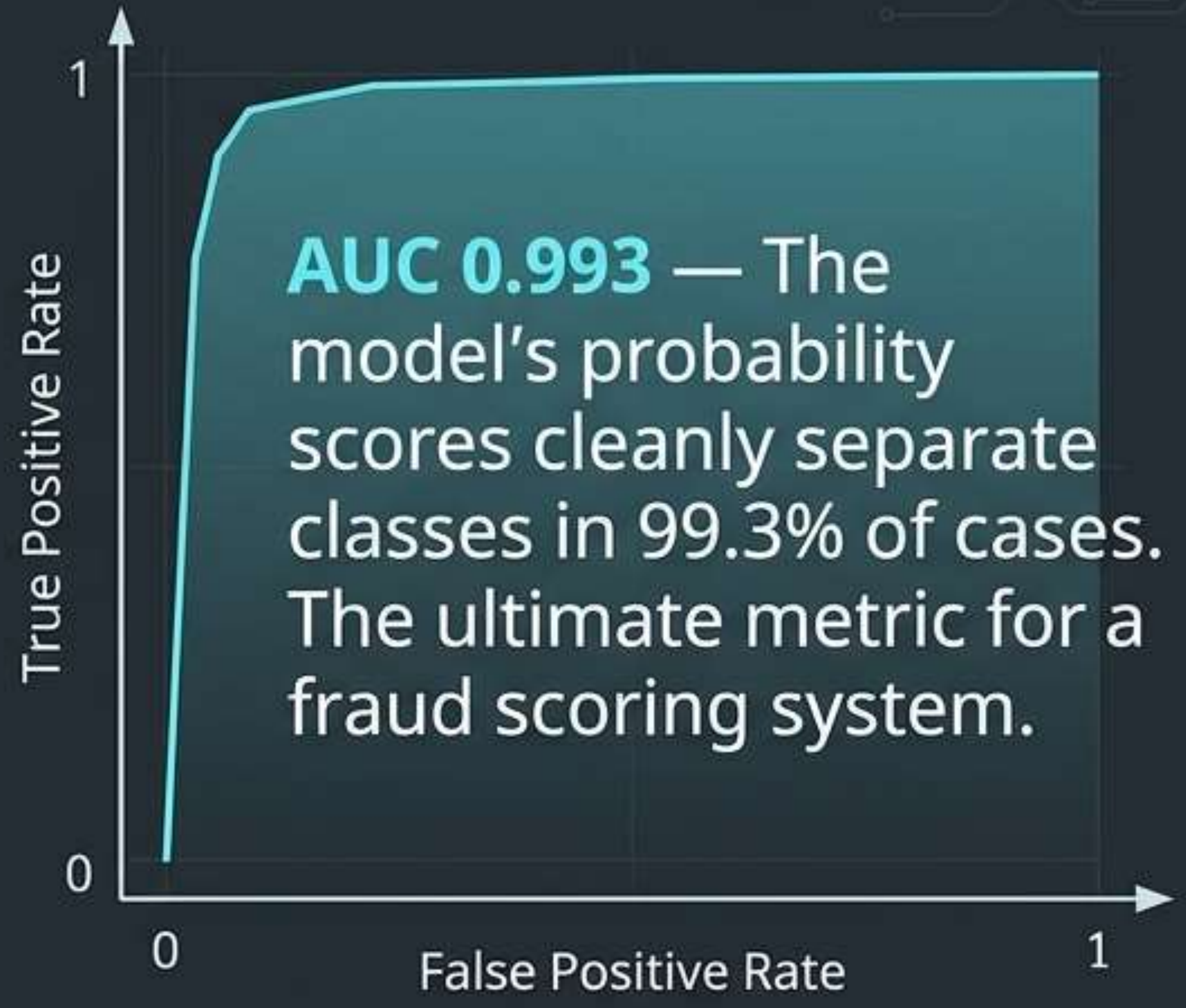
Key Innovation:

Trials are PRUNED unless Recall ≥ 0.89 **AND** Precision ≥ 0.93 simultaneously.

This prevents optimizing F1 at the expense of either metric.

Results — Near-Perfect AUC, High Precision

Metric	Target	Achieved	Status
F1 (fraud)	≥ 0.91	0.9069	Narrow miss (−0.3%)
Recall	≥ 0.89	0.8615	Miss (−2.9%)
Precision	≥ 0.93	0.9573	Met (+2.7%)
ROC-AUC	≥ 0.95	0.9930	Met (+4.3%)
MCC	–	0.8917	–



Threshold Calibration: Moving threshold from 0.50 → 0.87 added 3–4 F1 points at zero training cost.

Beyond Classification — 12-Tool Evidence Pipeline

Why a classifier isn't enough: It assigns probability, but it cannot verify external facts.

This produces human-readable evidence, not just a probability score.

Flask Web-App — Three Input Modes

Input Paths

1. **Paste text:** Job description entered directly.
2. **File upload:** PDF, DOCX, TXT, HTML, CSV.
3. **LinkedIn URL:** Scrapes and analyzes live posting.

What the User Sees

- Verdict banner (SAFE / **SUSPICIOUS** / **LIKELY_FAKE**) with confidence score.
- 12 tool result cards, each with 2–4 sentence LLM summary.
- Full LLM-written investigation report.

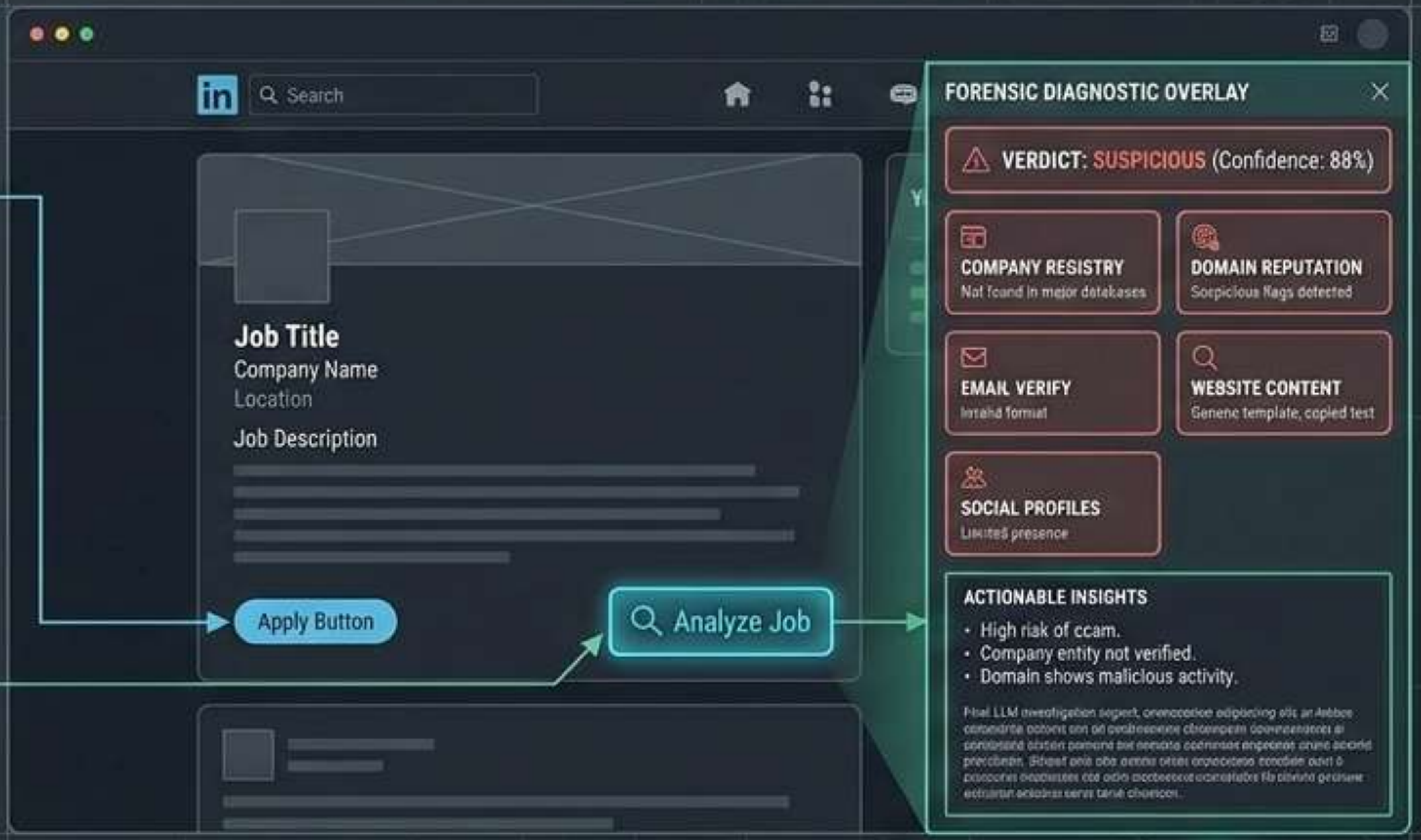
Real-Time Protection on LinkedIn

Bringing AI to the User

- Injects a floating “🔍 Analyze Job” button directly into the LinkedIn DOM.
- Powered by Google Gemini API and Vanilla JS — zero server latency.

The Experience

- **1-Click Execution**: Scrapes active job details directly from the webpage.
- **Instant Verdict:** Generates a color-coded overlay panel within 3–5 seconds.
- **Actionable Insights:** Translates complex fraud signals into simple bullet points for the job seeker.



Team Contributions

Arun Dutta

- Led data preprocessing pipeline development.
- Synthesized literature review & gap analysis.
- Synthesized results for final reporting.

Hritik Roshan Maurya

- Engineered Agentic AI & LangChain orchestrator.
- Built 16-field extraction logic.
- Implemented Web-App REST API.

COLLABORATIVE PROJECT EXECUTION

Vivek Bajaj

- Directed ML pipeline strategy.
- Performed Optuna HPO (25 trials).
- Calibrated thresholds & deployed to HuggingFace.

Vishwas Mehta

- Developed Chrome Extension & Browser Integration.
- Implemented Metadata Anomaly Detector.
- Led LinkedIn DOM scraping efforts.